

The sets E_n are disjoint, and since C is closed

$$F \setminus C = \{x \in F : \text{dist}(x, C) > 0\}$$

$$= \bigcup_{n=0}^{\infty} E_n$$

(3.4)

claim 2 If $n \geq 0$ then $\text{dist}(E_{2n}, E_{2n+2}) > 0$.

Indeed, let $x \in E_{2n}$, $y \in E_{2n+2}$ and we have

$$\frac{1}{2n+1} \leq \text{dist}(x, C) \leq d(x, y) + d(y, C) < d(x, y) + \frac{1}{2n+2},$$

($x \in E_{2n}$) ($y \in E_{2n+2}$)

and so $d(x, y) \geq \frac{1}{2n+1} - \frac{1}{2n+2} =: \eta_n > 0$. This proves claim 2.

As μ^* is a metric outer measure we obtain (by induction)

$$\forall n \in \mathbb{N} \quad \sum_{k=0}^n \mu^*(E_{2k}) = \mu^*\left(\bigcup_{k=0}^n E_{2k}\right) \leq \mu^*(F) < \infty$$

and, similarly, $\sum_{k=1}^n \mu^*(E_{2k-1}) \leq \mu^*(F) < \infty$.

We deduce that the series

$$\sum_{n=0}^{\infty} \mu^*(E_{2n}) \quad \text{and} \quad \sum_{n=1}^{\infty} \mu^*(E_{2n-1}) \quad \text{converge,}$$

and, in turn, $\sum_{n=0}^{\infty} \mu^*(E_n)$ converges. (3.5)

claim 3 For every $n \in \mathbb{N}$, $\text{dist}(F \setminus C, \bigcup_{n=0}^k E_n) > 0$.

Indeed, if $x \in F \setminus C$ and $y \in \bigcup_{n=0}^k E_n$, then

$$\frac{1}{k+1} \leq d(y, C) \leq d(x, y)$$

Again using the fact that μ^* is a metric outer measure, we have

$$\begin{aligned}
 \mu^*(F \cap C) + \mu^*(F \setminus C) &= \mu^*(F \cap C) + \mu^*\left(\bigcup_{n=0}^{\infty} E_n\right) \quad (\text{by (3.4)}) \\
 &= \mu^*(F \cap C) + \mu^*\left(\bigcup_{n=0}^k E_n \cup \bigcup_{n=k+1}^{\infty} E_n\right) \\
 \text{(subadditivity)} \quad &\leq \mu^*(F \cap C) + \mu^*\left(\bigcup_{n=0}^k E_n\right) + \mu^*\left(\bigcup_{n=k+1}^{\infty} E_n\right) \\
 \text{(subadditivity)} \quad &\leq \mu^*(F \cap C) + \mu^*\left(\bigcup_{n=0}^k E_n\right) + \sum_{n=k+1}^{\infty} \mu^*(E_n) \\
 \text{(claim 3)} \quad &= \mu^*\left(\underbrace{(F \cap C) \cup \bigcup_{n=0}^k E_n}_{=F}\right) + \sum_{n=k+1}^{\infty} \mu^*(E_n) \\
 \text{(monotonicity)} \quad &\leq \mu^*(F) + \sum_{n=k+1}^{\infty} \mu^*(E_n).
 \end{aligned}$$

letting $n \rightarrow \infty$ and using (3.5) we conclude that
 $\mu^*(F \cap C) + \mu^*(F \setminus C) \leq \mu^*(F)$,
 and this proves claim 1. $\blacksquare$

Proposition 3.13

The outer measure H^s_0 , $0 \leq s < \infty$ (see 2.4), and the outer measure $(\mathbb{L}^N)^*$ (see (2.2)) are metric outer measures, (i.e. $\text{dist}(E, F) > 0 \Rightarrow \mu^*(E \cup F) = \mu^*(E) + \mu^*(F)$)

In particular, every Borel set of $\mathbb{R}^N$ is H^s_0 -measurable and $(\mathbb{L}^N)^*$ -measurable.

and fact #4

It will be shown in thm 4.9 (Carathéodory) that this is in fact a σ -algebra

Def 3.14 The class of all $(\mathbb{L}^N)^*$ -measurable sets of $\mathbb{R}^N$ (see Def 3.1 and (3.3)) is called σ -algebra of Lebesgue measurable sets.

Proof of Proposition 3.13 we only prove it for $(\mathbb{L}^N)^*$.
 In view of Prop 3.12, if we show that $(\mathbb{L}^N)^*$ is a
 metric outer measure, then

$$\beta(\mathbb{R}^N) \subseteq \{ \text{Lebesgue measurable sets} \}.$$

let $E, F \subseteq \mathbb{R}^N$ be such that

$$d := \text{dist}(E, F) > 0.$$

claim $(\mathbb{L}^N)^*(E \cup F) \geq (\mathbb{L}^N)^*(E) + (\mathbb{L}^N)^*(F).$

Note that since $(\mathbb{L}^N)^*$ is subadditive, if the claim holds then

$$(\mathbb{L}^N)^*(E \cup F) = (\mathbb{L}^N)^*(E) + (\mathbb{L}^N)^*(F). \quad (\text{see Def 3.11})$$

Consider a sequence $\{R_n\}_{n \in \mathbb{N}}$ of rectangles such that

$$E \cup F \subseteq \bigcup_{n=1}^{\infty} R_n.$$

By partitioning each rectangle R_n into smaller rectangles,
 if necessary, we can assume that $\text{diam } R_n < \frac{d}{2} \quad \forall n \in \mathbb{N}.$

$$\text{Hence, if } R_n \cap E \neq \emptyset \Rightarrow R_n \cap F = \emptyset$$

$$R_n \cap F \neq \emptyset \Rightarrow R_n \cap E = \emptyset.$$

Therefore, we can divide the sequence $\{R_n\}$ into two sequences,
 one covering E , and the other covering F .

It follows that

$$\sum_{n=1}^{\infty} \ell(R_n) \geq \sum_{R_n \cap E \neq \emptyset} \ell(R_n) + \sum_{R_n \cap F \neq \emptyset} \ell(R_n)$$

$$\geq (\mathbb{L}^N)^*(E) + (\mathbb{L}^N)^*(F).$$

Taking the infimum over all sequences $\{R_n\}$ covering $E \cup F$

$$\text{give } (\mathbb{L}^N)^*(E \cup F) \geq (\mathbb{L}^N)^*(E) + (\mathbb{L}^N)^*(F),$$

proving the claim. ■

So we now have

$\mathcal{B}(\mathbb{R}^n) \subsetneq \{\text{Lebesgue measurable sets}\} \subsetneq \mathcal{P}(\mathbb{R}^n)$
we will see that both inclusions are strict.

Examples 3.15

a) $\{\text{Lebesgue measurable sets}\} \subsetneq \mathcal{P}(\mathbb{R}^n)$.

Recall the set $V \subset [-1,1]$, with $0 < (\mathcal{L}^1)^*(V) \leq (\mathcal{L}^1)^*([-1,1])$
(Rmk 2.13) + (Prop 2.11) = 2.

If V was Lebesgue measurable, because $(\mathcal{L}^1)^*$ is translation invariant (see Rmk 2.12) then

$r_k + V$ is Lebesgue measurable for all $k \in \mathbb{N}$, with these sets being mutually disjoint, so for every $n \in \mathbb{N}$

$$6 \geq (\mathcal{L}^1)^*\left(\underbrace{\bigcup_{k=1}^n (r_k + V)}_{\subset [-3,3]}\right) = \sum_{k=1}^n (\mathcal{L}^1)^*(r_k + V) = n (\mathcal{L}^1)^*(V),$$

and this is not possible for $n \gg 1$.

b) The Cantor set: an ^{Borel} uncountable set in $[0,1]$ with $(\mathcal{L}^1)^* = 0$.

In view of Prop 2.11, $(\mathcal{L}^1)^*({\{a\}}) = 0$ so if $E \subseteq \mathbb{R}$ is countable, by subadditivity $(\mathcal{L}^1)^*(E) = 0$.

The Cantor set $C \subseteq [0,1]$ is compact, uncountable, and $(\mathcal{L}^1)^*(C) = 0$.

Remark 3.13*

(i) In Def 0.1 we introduced the concept of Lebesgue measurability as:

$$\mu^*(A) + \mu^*(X \setminus A) = \mu^*(X) \quad (\text{with } \mu^* := (\mathcal{L}^1)^*, X = \mathbb{I})$$

(ii) In Def 3.1 we introduced the concept of measurability as
 $A \subseteq X$ is μ^* -measurable if $\forall F \subseteq X$

$$\mu^*(F \cap A) + \mu^*(F \setminus A) = \mu^*(F).$$

(iii) Hmw #1 (2): It μ^* is a regular outer measure then

$$(i) \Leftrightarrow (ii).$$

(iv) The Lebesgue outer measure is regular, hence

$$\text{Def 0.1} = \text{Def 3.1 for } (\mathcal{L}^1)^*.$$

Indeed, let $E \subseteq \mathbb{I}$. ? $\exists F \subseteq (\mathcal{L}^1)^*$ -measurable st.

$$E \subseteq F \text{ and } (\mathcal{L}^1)^*(E) = (\mathcal{L}^1)^*(F).$$

If $(\mathcal{L}^1)^*(E) = +\infty$, choose $F := \mathbb{R}$ (measurable by (i) or (ii)).

Suppose $(\mathcal{L}^1)^*(E) < +\infty$. $\forall k \in \mathbb{N}$ let $I_n^{(k)}$ be intervals st.

$$E \subseteq \bigcup_{n=1}^{\infty} I_n^{(k)}, \quad \sum_{n=1}^{\infty} \lambda(I_n^{(k)}) \leq (\mathcal{L}^1)^*(E) + 1/k$$

$$= (\mathcal{L}^1)^*(I_n^{(k)}) \text{ by Prop 2.11}$$

Prop 3.12 $\Rightarrow \bigcup_{n=1}^{\infty} I_n^{(k)}$ is Borel $\Rightarrow (\mathcal{L}^1)^*$ -measurable because

(Prop 3.13) $(\mathcal{L}^1)^*$ is a metric outer measure.

In turn, $F := \bigcap_{k=1}^{\infty} \left[\bigcup_{n=1}^{\infty} I_n^{(k)} \right] \in \beta(\mathbb{R})$, hence still $(\mathcal{L}^1)^*$ -meas.

clearly

$$E \subseteq F \text{ and } \forall k \quad (\mathcal{L}^1)^*(F) \leq (\mathcal{L}^1)^*\left(\bigcup_{n=1}^{\infty} I_n^{(k)}\right)$$

$$\leq (\mathcal{L}^1)^*(E) + 1/k,$$

$$\text{hence } (\mathcal{L}^1)^*(F) = (\mathcal{L}^1)^*(E).$$

(v) In Rmk 0.2 we claimed that $A \subset I$ (bounded interval) is Lebesgue measurable iff $\forall \varepsilon > 0 \exists A_\varepsilon$ "elementary set" (finite union of disjoint intervals) s.t.

$$(\mathbb{I}_\varepsilon)^*(A \Delta A_\varepsilon) = (\mathbb{I}_\varepsilon)^*((A \setminus A_\varepsilon) \cup (A_\varepsilon \setminus A)) < \varepsilon.$$

Indeed, we show that (ii) $\Rightarrow$ (v) and (v) $\Rightarrow$ (i)

with $\boxed{(i) \Leftrightarrow (ii)}$ by (iii) + (iv)

• (v) $\Rightarrow$ (i) let $\varepsilon > 0$ and find A_ε "elementary set" s.t.

$$(\mathbb{I}_\varepsilon)^*(A \Delta A_\varepsilon) < \varepsilon.$$

$$A \subseteq A_\varepsilon \cup (A \Delta A_\varepsilon), \quad I \setminus A \subseteq (I \setminus A_\varepsilon) \cup (A_\varepsilon \setminus A) \subseteq (I \setminus A_\varepsilon) \cup (A_\varepsilon \Delta A)$$

$$\text{so } (\mathbb{I}_\varepsilon)^*(A) + (\mathbb{I}_\varepsilon)^*(I \setminus A) \leq (\mathbb{I}_\varepsilon)^*(A_\varepsilon) + (\mathbb{I}_\varepsilon)^*(A_\varepsilon \Delta A) + 2\varepsilon$$

$$(A_\varepsilon \text{ measurable}) = (\mathbb{I}_\varepsilon)^*(I) + 2\varepsilon.$$

$$\text{let } \varepsilon \rightarrow 0^+ \text{ and get } (\mathbb{I}_\varepsilon)^*(A) + (\mathbb{I}_\varepsilon)^*(I \setminus A) \leq (\mathbb{I}_\varepsilon)^*(I). \\ (\geq \text{always})$$

• (ii) $\Rightarrow$ (v) let $\varepsilon > 0$ and choose E_n elementary sets (intervals)

$$\text{s.t. } A \subseteq \bigcup_{n=1}^{\infty} E_n, \quad \sum_{n=1}^{\infty} (\mathbb{I}_\varepsilon)^*(E_n) \leq (\mathbb{I}_\varepsilon)^*(A) + \varepsilon.$$

wlog write

$$\bigcup_{n=1}^{\infty} E_n =: A_\varepsilon \text{ disjoint union of}$$

finitely many intervals, and because $(\mathbb{I}_\varepsilon)^*(A) < \infty$ we can take $k > 1$ so that

$$\sum_{n=k+1}^{\infty} (\mathbb{I}_\varepsilon)^*(E_n) < \varepsilon.$$

we have

$$(\mathbb{I}_\varepsilon)^*(A \Delta A_\varepsilon) \leq (\mathbb{I}_\varepsilon)^*(A \setminus A_\varepsilon) + (\mathbb{I}_\varepsilon)^*(A_\varepsilon \setminus A)$$

$$\leq (\mathbb{I}_\varepsilon)^*\left(\bigcup_{n=k+1}^{\infty} E_n\right)$$

$$\leq \sum_{n=k+1}^{\infty} (\mathbb{I}_\varepsilon)^*(E_n) < \varepsilon$$

$$< 2\varepsilon.$$

$$\downarrow \text{ " } (A \text{ satisfies (ii)}) \\ \leq (\mathbb{I}_\varepsilon)^*(\bigcup_{n=1}^{\infty} E_n \setminus A)$$

$$= (\mathbb{I}_\varepsilon)^*(\bigcup_{n=1}^{\infty} E_n) - (\mathbb{I}_\varepsilon)^*(\bigcup_{n=1}^{\infty} E_n \cap A)$$

$$= (\mathbb{I}_\varepsilon)^*(\bigcup_{n=1}^{\infty} E_n) - (\mathbb{I}_\varepsilon)^*(A)$$

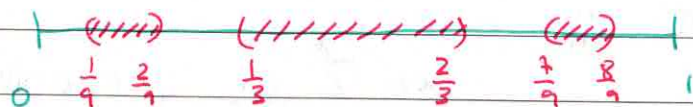
$$\leq \sum_{n=1}^{\infty} (\mathbb{I}_\varepsilon)^*(E_n) - (\mathbb{I}_\varepsilon)^*(A) < \varepsilon.$$

Divide $[0,1]$ into three equal subintervals and remove the middle third $I_1 := (\frac{1}{3}, \frac{2}{3})$.



Divide each of the two closed intervals $[0, \frac{1}{3}]$, $[\frac{2}{3}, 1]$ into 3 equal subintervals and remove the middle ones

$$I_{12} := (\frac{1}{3^2}, \frac{2}{3^2}), \quad I_{22} := (\frac{7}{3^2}, \frac{8}{3^2}).$$



Continue at each step n removing 2^{n-1} middle third intervals $I_{1n}, \dots, I_{2^{n-1}n}$ of length $\frac{1}{3^n}$.

The Cantor set C is defined as

$$C := [0,1] \setminus \left(\bigcup_{n=1}^{\infty} \bigcup_{k=1}^{2^{n-1}} I_{kn} \right).$$

C is closed, because its complement is a family of open intervals and because $[0,1]$ and C are Borel sets, hence $(\mathcal{L}^1)^*$ measurable (see Prop 3.13)

$$1 = \mathcal{L}([0,1]) = \underset{\uparrow \text{Prop 2.11}}{(\mathcal{L}^1)^*}([0,1]) = (\mathcal{L}^1)^*([0,1] \setminus C \cup C)$$

$$= (\mathcal{L}^1)^*([0,1] \setminus C) + (\mathcal{L}^1)^*(C), \text{ so}$$

$$(\mathcal{L}^1)^*(C) = 1 - (\mathcal{L}^1)^*\left(\bigcup_{n=1}^{\infty} \bigcup_{k=1}^{2^{n-1}} I_{kn}\right) =$$

↓
disjoint union of intervals,
hence Lebesgue measurable

$$\begin{aligned}
&= 1 - \sum_{n=1}^{\infty} \sum_{k=1}^{2^{n-1}} \lambda(I_{kn}) \\
&= 1 - \sum_{n=1}^{\infty} \sum_{k=1}^{2^{n-1}} \frac{1}{3^n} = 1 - \sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n} \\
&= 1 - \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = 1 - \frac{1}{2} \frac{2/3}{1-2/3} \\
&= 1 - \frac{1}{2} \cdot 2 = 0.
\end{aligned}$$

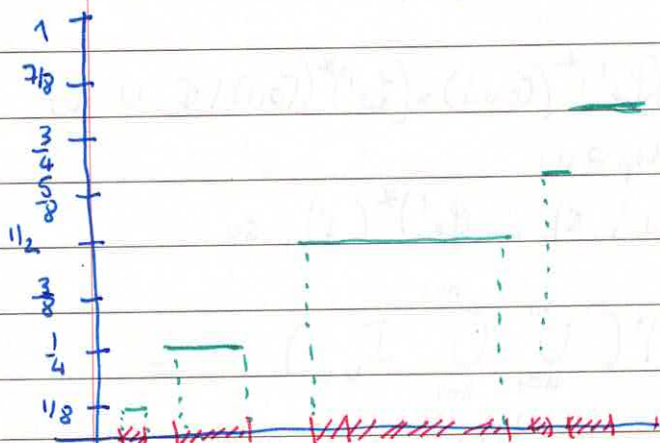
To prove that $\mathbb{C}$ is uncountable, it can be shown that $x \in \mathbb{C}$ iff

$$x = \sum_{n=1}^{\infty} \frac{c_n}{3^n}, \quad c_n \in \{0, 2\}.$$

let $f: \mathbb{C} \rightarrow [0, 1]$, $f(x) := \sum_{n=1}^{\infty} \frac{d_n}{2^n}$ with

$$d_n := \begin{cases} 1 & \text{if } c_n = 2, \\ 0 & \text{if } c_n = 0. \end{cases}$$

Cantor Function It can be shown that f is increasing, and has the same values at the endpoints of each removed interval I_{kn} , thus f can be extended to a continuous function on $[0, 1]$. This function is called the **Cantor function** (3.6)



$f(\mathbb{C}) = [0, 1]$, so $\mathbb{C}$ has to be uncountable.

Note that $f' = 0$ in I_{kn} so $f' = 0$ except at most on $\mathbb{C}$, a set of $(\mathbb{R}^1)^*$ measure zero.

(c) We will see later on that $\beta(\mathbb{R}^n) \subsetneq \{\text{Lebesgue measurable sets}\}$.
(see example 4.12) (A.12)

4. MEASURES

Def 4.1 Let $(X, \mathcal{M})$ be a measurable space. A measure on $(X, \mathcal{M})$ is a function $\mu: \mathcal{M} \rightarrow [0, +\infty]$ such that

(i) $\mu(\emptyset) = 0$,

(ii) (countably additive) $\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n)$ for every family $\{E_n\} \in \mathcal{M}$ of pairwise disjoint sets.

$(X, \mathcal{M}, \mu)$ is called a measure space.

Examples 4.2

a) Counting measure on a set X , with $\mathcal{M} = \mathcal{P}(X)$:

$$\mu(E) = \begin{cases} \#E & \text{if } E \text{ is finite, } (\# \text{ of elems. of } E) \\ +\infty & \text{---} \end{cases}$$

b) Dirac measure. Let $(X, \mathcal{M})$ be a measurable space, and let $a_0 \in X$.

$$\delta_{a_0}(E) := \begin{cases} 1 & \text{if } a_0 \in E, \\ 0 & \text{if } a_0 \notin E, \end{cases} \quad E \in \mathcal{M}.$$

c) Let $X \neq \emptyset$ and consider the σ -algebra $\mathcal{M}$ of all subsets of X that are either countable or have a countable complement in X . Set

$$\mu(E) := \begin{cases} 0 & \text{if } E \text{ is countable,} \\ 1 & \text{if } E \text{ is uncountable,} \end{cases} \quad E \in \mathcal{M}.$$

Proposition 4.3

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $E, F \in \mathcal{M}$, $E \subseteq F$.

(i) $\mu(E) \leq \mu(F)$ [Monotonicity]

(ii) $\mu(F \setminus E) = \mu(F) - \mu(E)$ provided that $\mu(E) < \infty$.

Proof Because $F = E \cup \underbrace{(F \setminus E)}_{\text{disjoint}}$, disjoint union of measurable sets, we have

$$\mu(F) = \mu(E) + \mu(F \setminus E) \geq \mu(E),$$

proving (i). Furthermore, if $\mu(E) < \infty$ then we get

$$\mu(F \setminus E) = \mu(F) - \mu(E). \quad \blacksquare$$

Proposition 4.4

Let $(X, \mathcal{M}, \mu)$ be a measure space, let $E, F \in \mathcal{M}$ with $\mu(E \cap F) < \infty$. Then

$$\mu(E \cup F) = \mu(E) + \mu(F) - \mu(E \cap F).$$

Proof Write

$$E \cup F = E \setminus (E \cap F) \cup F \setminus (E \cap F) \cup (E \cap F),$$

a union of three disjoint measurable sets.

$$\text{Then } \mu(E \cup F) = \mu(E \setminus (E \cap F)) + \mu(F \setminus (E \cap F)) + \mu(E \cap F)$$

$$\begin{aligned} \text{(Prop. 4.3, ii)} \quad &= \mu(E) - \mu(E \cap F) + \mu(F) - \mu(E \cap F) + \mu(E \cap F) \\ &= \mu(E) + \mu(F) - \mu(E \cap F). \end{aligned} \quad \blacksquare$$

Proposition 4.5 [Countable subadditivity]

Let $(X, \mathcal{M}, \mu)$ be a measure space, and let $\{E_n\}_{n \in \mathbb{N}} \subseteq \mathcal{M}$.

Then
$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} \mu(E_n).$$

Proof Write $E'_n := E_n \setminus \bigcup_{1 \leq k < n} E_k$, $n \geq 2$, $E'_1 := E_1$.

Then $E'_n \in \mathcal{M}$ and are mutually disjoint, hence

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \mu\left(\bigcup_{n=1}^{\infty} E'_n\right) = \sum_{n=1}^{\infty} \mu(E'_n) \leq \sum_{n=1}^{\infty} \mu(E_n).$$

Prop 4.3 (i) $\blacksquare$

end of lect #5

Proposition 4.6 [Measure of an Increasing Union]

Let $(X, \mathcal{M}, \mu)$ be a measure space, and let $E_n \in \mathcal{M}$ be such that $E_n \subseteq E_{n+1}$, $\forall n \in \mathbb{N}$. Then

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \mu(E_n)$$

(see also Prop 6.2, still valid for regular outer measures).

Proof If $\mu(E_n) = +\infty$ for some $n \in \mathbb{N}$, then by monotonicity both sides are $= \infty$.

Assume that $\mu(E_n) < +\infty$ for all $n \in \mathbb{N}$.

Then
$$\bigcup_{n=1}^{\infty} E_n = \bigcup_{n=1}^{\infty} (E_n \setminus E_{n-1})$$
 where $E_0 := \emptyset$.

The right hand side is a disjoint union of measurable sets, so

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n \setminus E_{n-1})$$

$$= \lim_{K \rightarrow \infty} \sum_{n=1}^K \mu(E_n \setminus E_{n-1})$$

$$\begin{aligned} & \text{(Prop 4.3 (ii))} \\ & \text{because } \mu(E_{n-1}) < +\infty \end{aligned} = \lim_{K \rightarrow \infty} \sum_{n=1}^K [\mu(E_n) - \mu(E_{n-1})]$$

$$= \lim_{K \rightarrow \infty} \mu(E_K) \quad (\text{used } \mu(\emptyset) = 0). \quad \blacksquare$$

Proposition 4.7 [Measure of a Decreasing Intersection]

Let $(X, \mathcal{M}, \mu)$ be a measure space, and let $E_n \in \mathcal{M}$ be such that

$$E_n \supseteq E_{n+1}, \quad \forall n \in \mathbb{N}, \quad \text{and} \quad \mu(E_1) < \infty.$$

Then
$$\mu\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \mu(E_n).$$

Proof By De Morgan's law

$$E_1 \setminus \bigcap_{n=1}^{\infty} E_n = \bigcup_{n=1}^{\infty} (E_1 \setminus E_n), \quad \text{with}$$

$$E_1 \setminus E_1 \subseteq E_1 \setminus E_2 \subseteq \dots \subseteq E_1 \setminus E_n \subseteq \dots$$

By Prop 4.6.,

$$\mu(E_1 \setminus \bigcap_{n=1}^{\infty} E_n) = \lim_{n \rightarrow \infty} \mu(E_1 \setminus E_n), \quad (*)$$

and by monotonicity and the fact that $\mu(E_1) < \infty$, we have

$$\mu\left(\bigcap_{n=1}^{\infty} E_n\right), \quad \mu(E_n) < \infty \quad \text{for all } n \in \mathbb{N}.$$

Therefore, by Prop 4.3 (ii) (*) can be written as

$$\mu(E_1) - \mu\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} [\mu(E_1) - \mu(E_n)],$$

and so

$$\mu\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \mu(E_n). \quad \blacksquare$$

Def 4.8 Given a measure space $(X, \mathcal{M}, \mu)$, the measure

μ is said to be complete if for every $E \in \mathcal{M}$ with $\mu(E) = 0$ it follows that every $F \subseteq E$ also belongs to $\mathcal{M}$.

[equivalently, we say that $\mathcal{M}$ is complete with respect to μ].